

PIC: Domain Generalization by Path Information Constraint

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Abstract

Despite the widespread use of deep neural networks in various tasks, their weak generalization ability to out-of-distribution data has always been one of the challenges hindering their practical applications. Domain generalization (DG) aims to address this issue by learning domain-robust features across source domains, enabling models to generalize well to unseen target domains. Although DG has received extensive attention, existing solutions often overlook the relationships between different source domains and fail to make good utilization of the information contained in the source domain data to enhance the model's generalization ability. In this paper, we propose the Path Information Constraint (PIC) method that achieves diversified exploration of the training process by constructing different learning paths in source domains. Experimental results show that our method achieves competitive results on multiple challenging benchmarks. PIC outperforms the best non-ensembled method by an average of 0.9% across four benchmarks. Furthermore, ensembling PIC by averaging multiple independently trained models leads to further improvements.

1. Introduction

Deep neural networks have demonstrated remarkable performance in a range of studies. However, their success depends on the assumption of independent and identically distributed (i.i.d.) data between training and test sets. When encountering data from an unknown and different distribution, their performance will drop drastically. In practical applications, different deployment environments often bring different data distributions, but it is impractical to annotate enough training data for each deployment task. Labeling data is a time-consuming process that requires a lot of expertise. Additionally, for privacy and data security reasons, it is sometimes strictly forbidden to collect data for deployment scenarios. The unreliability of models caused by data distribution hinders the application of machine learning. Therefore, achieving model robustness in out-of-distribution (OOD) scenarios is becoming increasingly important in practical machine learning applications.

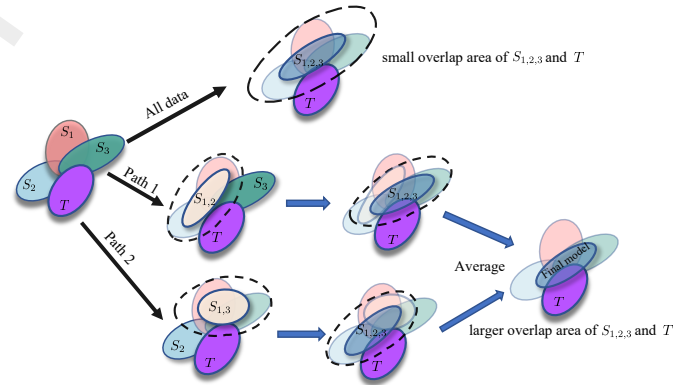


Figure 1: The illustration of path information constraint. The distributions of 3 source domains and 1 target domain are shown in the four left areas, where the target domain distribution is unknown during training. The common case of processing data together is shown in the All data path, and the learning process of different paths is shown in Path 1 and Path 2. The dashed area is the distribution of training data and the center of it is the mapping of trained model. The models of all paths are averaged to get the final model. The overlap between the final model and the target domain represents the generalization ability in the target domain. Multiple paths can provide diversity in training, facilitate the model to learn robust features, and ultimately obtain a larger overlap.

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To achieve the model's robustness, researchers have proposed various learning paradigms, such as Domain Adaptation (DA) [1, 2, 3] and Domain Generalization (DG) [4, 5, 6]. Different from Domain Adaptation, which can access unlabeled target domain data during training, Domain Generalization can only access source domain data. The unknownness of target domain data brings great challenges to domain generalization. But it also makes Domain Generalization important in the deployment tasks of unseen application scenarios.

In domain generalization tasks, most existing methods handle data from all source domains together and align feature representations to learn domain-invariant features. However, these methods mostly treat all learning tasks across different domains as a whole, ignoring the differences between tasks from different domains. This results in most methods [7, 8, 9] only utilizing class information, but ignoring the domain information contained in the data. Even in methods that utilize domain information (such as DANN [10] and CDANN [11]), they only treat domain information as a form of a supervisory signal, overlooking the impact of different domain data distributions on the diversity of model training.

In this work, we propose a path information constraint (PIC) method that utilizes the domain information of source domains data to construct multiple learning paths and explore diversity through path variation during the training process. We believe that whether learners initially learn about elephants and horses in cartoons or paintings, it will ultimately help them in distinguishing between these animals in the real world. By learning through different paths, learners can achieve diverse intellectual development effects. During training, we construct multiple learning paths, save the learning outcomes of each path, and supervise the training process in the next path. A step-by-step strategy is used to fit the data distribution in every single path. Since the data distribution changes differently in each path, multiple paths can increase training diversity and ultimately learn robust cross-domain feature representations.

Our motivation for proposing the path constraint is shown in Figure 1. In the figure, the source domain data are processed

together or on two paths. The representations of the model is distributed within the dashed area. The mapping after training is shown in the center of the dashed line. As the training steps increases, the feature map region gradually shrinks to the middle area. The model is more and more fitted to the distribution of the training data, and the model's discriminative ability is enhanced. The regions $S_{1,2,3}$ represent the final mapping distribution. The larger the overlap between the final mapping and the target domain, the stronger the generalization ability achieved. Compared with processing data together, training with the step-by-step strategy fits the data distribution multiple times, and each fitting updates the intermediate domain. In addition, the models obtained from different paths should have similar information. Thus the information learned in previous paths can guide the learning direction of the next path, preventing generalization disasters caused by unreasonable path design. Finally, we average all the information learned from different paths to obtain the final training results.

We propose a simple implementation of path information constraints. We train a base model on one path that simulates the step-by-step learning process of humans. Then, we construct a new learning path and train it under the supervision of the previous path information. **By setting the task paths, PIC increases the diversity of the training process. And the final model obtained by averaging multiple diversity models can have a larger overlap with the target domain.**

By combining with ensemble learning, our method can achieve better performance. In common ensemble learning methods, Stochastic Weight Averaging Densely (SWAD) [12] requires the frequent computation of performance on the validation set to determine the starting and stopping times of iterations, resulting in high computational costs. Ensemble of Averages (EoA) [13] is experimental diversity relies on multiple independent experiments, which ultimately need to be averaged across a certain amount of experiments. We further combine PIC with ensemble learning by averaging multiple independently trained models. Experiments show that the ensemble of PIC (EPIC) can achieve even better generalization perfor-

mance. In contrast, since the training process of PIC has already explored diversity through paths, the EPIC only requires simple averaging of a small amount of independently trained models to achieve excellent performance.

Our main contributions are as follows:

- Our method is simple and effective in both implementation and concept. The model can leverage the constraints from previous paths to learn more powerful features.
- We design a novel learning framework that mimics human learning patterns through progressive learning tasks and path information constrained to reduce the interference caused by changes in data distribution.
- Compared to common ensemble methods, EPIC does not need to frequently calculate on the validation set, and only needs a small number of independent training processes to achieve good results, reducing the computational overhead in the ensemble learning process.
- The experimental results show that PIC achieves competitive performance and steady improvement on a series of challenging domain generalization benchmarks.

2. Related Work

2.1. Domain Generalization

Domain generalization process labeled data from source domains, and the goal is to extract features that are robust to changes in the data distribution and generalize the learned features to unseen target domains. Over the years, significant progress has been made in domain generalization research, and researchers have proposed various algorithms. There are many meaningful works in domain generalization, which can be roughly divided into three categories: data augmentation, domain-invariant feature learning, and learning strategies.

(1) Data augmentation. This type of methods [14, 15, 16, 17] is the most intuitive way to improve the generalization ability of the model from the perspective of data. Nam et al. [18] proposed to preserve image content and randomize image styles in data augmentation to reduce style differences between samples.

Yao et al. [19] obtained excellent results by selecting proxy samples with a suitable learning difficulty to train the model.

(2) Domain-invariant feature learning. It has always been the focus and key technique of domain generalization. It enables the model to generalize well to different domains by learning domain-invariant features. There is a lot of excellent works in domain-invariant feature learning research. Several works [20, 21, 22] learned a domain-invariant feature representation such that the best linear classifier based on this representation can match well across domains. Some works [7, 23, 24] used the kernel function, which utilizes a series of statistics to align features from different domains.

(3) Learning strategies. It utilizes different learning methods to improve the generalization ability of the model. Some works [25, 26, 27] used a meta-learning approach to learn generalization representations by simulating domain shift via constructing meta-learning tasks. Several works [28, 29, 30] focused on learning and optimizing the error-case distribution of the source domain, making the model robust. IDCL [31] has also achieved good performance by setting up cross-domain courses on the training paths to gradually reduce the migration difficulties caused by too much difference in data distribution. Unlike IDCL, we not only focus on different training paths but also the constraints of previous paths on the current path. Through different paths, the model can increase the diversity of data distribution changes during training. This helps to learn features that are stable to data shifts.

2.2. Adversarial learning in generalizability research

Our work is motivated by the advances in adversarial learning, which has played an important role in generalization research. Similar to the generator and discriminator in Generative Adversarial Network (GAN [32]), in generalization research [33, 34], the generator is used to extract sample features, and the discriminator is used to determine which domain the features come from. When the extracted features can confuse the discriminator's judgment, the domain-invariant features will be obtained. Ganin et al. proposed the Domain-Adversarial Neu-

ral Network (DANN [10]) for domain adaptation, which fully utilizes this idea and trains the generator and discriminator in an adversarial manner. Furthermore, in the research of adversarial feature disentanglement [35, 36], researchers separate semantic features (task-related) and stylistic features (domain-related), to obtain strong generalization performance in new domains.

2.3. Ensemble Learning

Ensemble learning methodologies leverage the strengths of multiple models to generate predictions and integrate their outputs using techniques such as boosting and mean aggregation to attain superior performance relative to a single model. The crux of these techniques lies in developing a range of diverse base models, which are then integrated using ensemble methods. This approach enables ensemble learning to optimize performance through a blend of diverse models. In domain generalization research, domain-specific expert models [37, 7] are set up. Each model is only an expert in its domain and not in other domains. Finally, different expert models are aggregated to achieve performance improvement.

In addition to aggregating different domain expert models, there are also works that consider model ensembling during training. Model Averaging (MA) obtains the average model parameters during training. Based on MA, the stochastic weight averaging (SWA [38]) algorithm does not average at the beginning but after a certain number of iterations until the end of training. SWAD [12] calculates the loss threshold to determine the start and end time of averaging, avoiding the model fitting into a sharp local minimum area while also increasing computational costs on the validation set. EoA [13] uses multiple sets of different hyperparameters and random seeds for independent training, generating diversity in training through randomness, and finally averaging multiple models separately, further reducing frequent calculations on the validation set.

3. Methodology

3.1. Symbol Definition

$\mathcal{D}_S$ denotes the data in source domains, which can be obtained during training, and $\mathcal{D}_T$ denotes the data in unseen tar-

get domains. $\mathcal{D}_S$ and $\mathcal{D}_T$ share a common label space, but have different data distributions. $\mathcal{D}_S = \{\mathcal{S}_1, \dots, \mathcal{S}_{N_d}\}$ have C classes and N_d source domains. In each source domain, samples are drawn from $\mathcal{S}_k = \{x_i^k, y_i^k\}_{i=1}^{N_k}$ where $y_i^k \in \{1, \dots, C\}$ and N_k is the sample size of $\mathcal{S}_k$.

3.2. Overall Structure

Motivated by adversarial learning, we chose the adversarial feature disentanglement algorithm as our adversarial learning method in the experiments and built upon it to incorporate Path Information Constraint (PIC). The Gradually Vanishing Bridge (GVB) method proposed by Cui et al. [39] has achieved outstanding performance on multiple benchmarks in domain adaptation tasks. Therefore, we choose to modify the GVB method to be competent for the DG scenario. The adversarial learning method consists of a feature extraction network G , a classifier network C , a feature disentanglement network F , and a discriminator network D . The network G can extract overall features f from the samples. The network F is used to obtain domain-related features γ from the features f . The features γ can capture the differences between current domain and intermediate domain. On the other hand, the classifier network generates domain-invariant features r after removing domain-related features. The discriminator is used to distinguish which domain the input features come from.

The stepwise learning and path information constraint are explained in Figure 2. In Path 1, we first take $\mathcal{S}_1$ and $\mathcal{S}_2$ to obtain their intermediate domain $\mathcal{S}_{1,2}$. Then, we use the intermediate domain $\mathcal{S}_{1,2}$ and domain $\mathcal{S}_3$ to train the model and obtain the intermediate domain $\mathcal{S}_{1,2,3}$ for all source domains. The best model on the validation dataset is selected to supervise the training of Path 2. In Path 2, $\mathcal{S}_1$ and $\mathcal{S}_3$ are used to obtain their intermediate domain $\mathcal{S}_{1,3}$, and then $\mathcal{S}_{1,3}$ and $\mathcal{S}_2$ are used to extract the intermediate domain $\mathcal{S}_{1,2,3}$.

Unlike Path 1, the training of Path 2 is further supervised by Path 1. We consider the model with the best performance on the validation dataset along a path as the best model for that path. After training, the best models of Path 1 and Path 2 are averaged

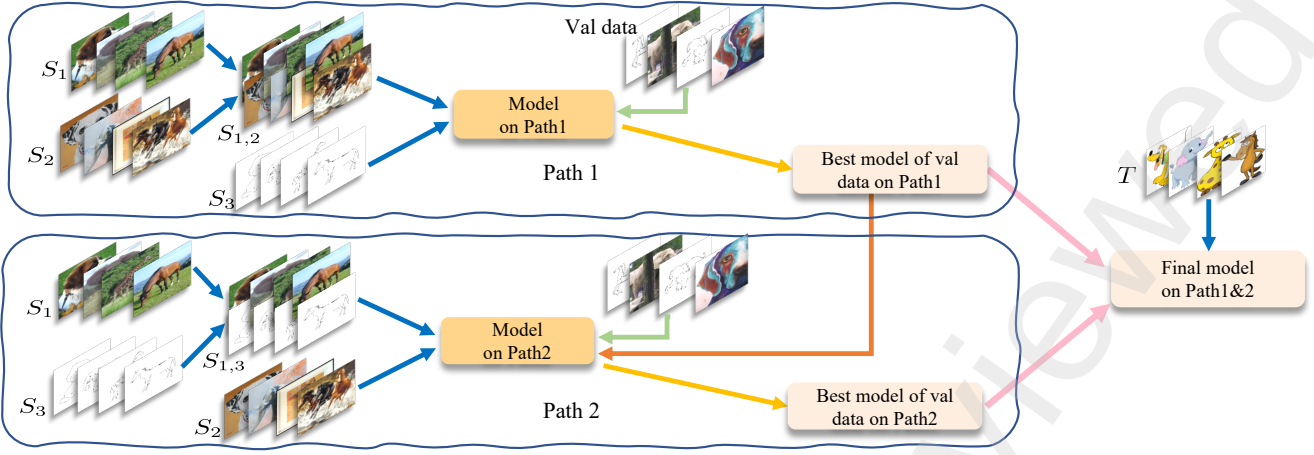


Figure 2: Illustration of stepwise learning and path information constraint. For convenience, the proposed method is explained in two paths. First, we train the model on Path 1 and save the best model on the validation data. Then use the saved model to constrain the learning process of Path 2. the best models of Path 1 and Path 2 are averaged to obtain the final PIC model. Finally, we test the performance of the model on the unseen target domain.

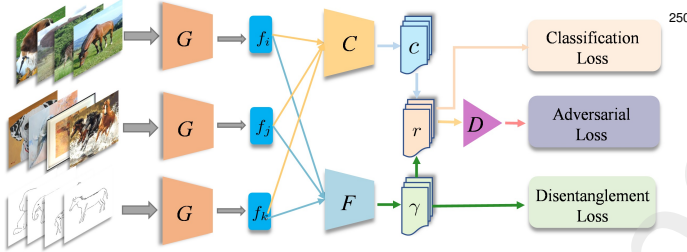


Figure 3: Structure of the designed adversarial learning on one path. The network is shared across different domains. G is a feature extraction network, which can extract the overall features f of different source domain data. F is the feature disentangling network, which can disentangle the domain-related features γ from f . C is the classifier network which can generate category responses features c , but c contain domain-related information. After subtracting γ from c , the domain-independent features r for classification can be obtained. Classification Loss, Adversarial Loss, and Disentanglement Loss are adopted during the training process on one path.

to obtain the final PIC model. In ensemble learning process, the above training process is followed by various random initialization of parameters to obtain multiple independently trained models, which are ensembled to obtain the final EPIC model.

3.3. Adversarial learning method

In the adversarial learning method, we adopted the Gradually Vanishing Bridge (GVB) auxiliary learning structure, which can serve as a bridge between different domains and measure the differences between domains and intermediate domains, ef-

fectively reducing the negative impact of domain-specific features and improving the generalization ability of the network. We first randomly select two source domains for training, interpolate the features of the two domains to obtain the intermediate domain features, and continuously update the intermediate domain features with the remaining source domains. Finally, the intermediate domain features of the entire source domain can be obtained. The design of the adversarial learning structure is illustrated in Figure 3.

All network parameters in the method are shared across different domains. The method first selects two source domains $\mathcal{S}_i$ and $\mathcal{S}_j$ to obtain features $f_i = G(x^i)$, $c_i = C(f_i)$, $\gamma_i = F(f_i)$ and $f_j = G(x^j)$, $c_j = C(f_j)$, $\gamma_j = F(f_j)$ where $x^i \in \mathcal{S}_i$ and $x^j \in \mathcal{S}_j$. Then, the data is interpolated to obtain features $f_{[i,j]}$, $c_{[i,j]}$, and $\gamma_{[i,j]}$ of the intermediate domain $\mathcal{S}_{[i,j]}$. The remaining source domain features f' , c' , γ' in turn to update the intermediate domain features, and finally obtain the overall intermediate features $f_{\{1,\dots,N_d\}}$, $c_{\{1,\dots,N_d\}}$ and $\gamma_{\{1,\dots,N_d\}}$. Less domain-specific features r can be generated from c and γ via Equation 1.

$$r = c - \gamma \quad (1)$$

The final overall intermediate-generated features $r_{\{1,\dots,N_d\}}$ are strong generalization features. The structure of the adversarial learning method is illustrated in Figure 3.

The losses adopted in the method include the classification

loss ℓ_{ce} of samples, the adversarial loss ℓ_{adv} generated by the discriminator network D and the loss ℓ_{dis} generated by the feature disentanglement network F .

The first is the most basic classification loss, which is used to guide the model through the classification task and is calculated in the form shown in Equation 2.

$$\ell_{ce} = \frac{1}{B} \sum_{i=1}^B \sum_{c=1}^C -y_{i,c} \log \frac{\exp(r_{i,c})}{\sum_{j=1}^C \exp(r_{i,j})} \quad (2)$$

where B represents the number of training samples in a training batch.

Then there is the adversarial loss adopted during the information disentangling process, which is used to guide the discriminator to identify whether the current sample is from an intermediate domain or a specific source domain. The expression for the loss is shown in Equation 3.

$$\begin{aligned} \ell_{adv} = & -\frac{1}{N_m} \sum_{i=1}^{N_m} w_m \log D(r_i) \\ & -\frac{1}{N_n} \sum_{i=1}^{N_n} (1 - w_m) \log(1 - D(r_i)), \end{aligned} \quad (3)$$

where $D(r_i)$ is the output of discriminator over r_i and $w_m = Ent(m)/(Ent(m) + Ent(n))$ is a weight used to balance the two parts of ℓ_{adv} . $Ent()$ is the mean entropy over samples which can be calculated as $Ent(*) = -\frac{1}{N_*} \sum_{i=1}^{N_*} \sum_{j=1}^C \frac{\exp(r_{i,j})}{\sum_{j=1}^C \exp(r_{i,j})} \log(\frac{\exp(r_{i,j})}{\sum_{j=1}^C \exp(r_{i,j})})$.

Finally, there is the disentanglement loss. We want the disentangled domain-related features to have a smaller and smaller share of the training process. That is, most of the features should be task-related and the L1 norm is used to minimize domain-related features γ . The loss expression is given by Equation 4.

$$\ell_{dis} = \sum_{i=1}^N \|\gamma_i\|_1 \quad (4)$$

3.4. Path Information Constraint

Existing adversarial feature disentanglement methods tend to process data all in one, ignoring the information on data variations between source domains. We propose a path information

constraint method to leverage the hidden data distribution diversity within source domains and learn features that are robust to shifts in data. For different learning paths, we first design a learning path on the source domain, save the best-performing model on this path, and load it as the supervisory model when training the network on the second path. Then we compute the differences between the supervisory model and the current training model. By designing different paths, different exploration directions can be provided. The previously learned knowledge on the previous path can constrain and guide the optimization of the current model, reducing the negative effects caused by inappropriate paths.

Compared to the adversarial learning method, the PIC architecture has an extra supervisory loss generated by the previous path. The supervisory loss is defined by the MMD-based regularization term. Given the mapping distribution P_{pre} generated by the previous path and P_{cur} generated by the current path under the same batch of data. It is proved [40, 7] that any probability distribution P is uniquely represented by an element in the reproducing kernel Hilbert space (RKHS) through the mean mapping. In practice, a kernel function is induced to mean map the distributions and calculate their mean values in RKHS $\mathcal{H}$. The calculation is shown in Equation 5.

$$\mu_{\mathbb{P}} := \mu(\mathbb{P}) = \mathbb{E}_{x \sim \mathbb{P}}[\phi(x)] = \mathbb{E}_{x \sim \mathbb{P}}[k(x, \cdot)] \quad (5)$$

where $k(x, \cdot)$ is the kernel function. Here we use the RBF kernel, $k(x, x') = \exp(-\frac{1}{2\sigma^2} \|x - x'\|^2)$, where σ is the bandwidth parameter. Based on MMD theory, we can measure the distance between P_{pre} and P_{cur} . The practical application of MMD uses unbiased estimation to calculate $\mu_{\mathbb{P}}$. Thus the distance between paths can be calculated as in formula 6.

$$\begin{aligned} MMD(P_{pre}, P_{cur}) = & \left\| \frac{1}{n_{pre}} \sum_{i=1}^{n_{pre}} k(P_{pre_i}, P_{pre'_i}) - \frac{1}{n_{cur}} \sum_{i=1}^{n_{cur}} k(P_{cur_j}, P_{cur'_j}) \right\|_{\mathcal{H}}^2 \\ = & \frac{1}{n_{pre}^2} \sum_{i=1}^{n_{pre}} \sum_{i'=1}^{n_{pre}} k(p_{pre_i}, p_{pre'_{i'}}) \\ & + \frac{1}{n_{cur}^2} \sum_{j=1}^{n_{cur}} \sum_{j'=1}^{n_{cur}} k(p_{cur_j}, p_{cur'_{j'}}) \\ & - \frac{2}{n_{pre}n_{cur}} \sum_{i=1}^{n_{cur}} \sum_{j=1}^{n_{pre}} k(p_{pre_i}, p_{cur_j}) \end{aligned} \quad (6)$$

Since the knowledge learned by different paths should be similar, the path information constraint can be implemented by minimizing that distance $\ell_{pic} = MMD(P_{pre}, P_{cur})$.

In summary, the overall loss is shown in Equation 7.

$$\ell_{total} = \ell_{ce} + \ell_{dis} + \ell_{adv} + \alpha \ell_{pic} \quad (7)$$

We set the weight α for ℓ_{pic} to adjust the constraint ability of different paths. More details about path construction and parameter setting will be presented in the experimental section.

4. Experiments

In this section, we will introduce the benchmark dataset used in the experiments in Sec 4.1, the experimental details in Sec 4.2, the comparison of experimental results in Sec 4.3, the results analysis in Sec 4.4, and the ablation study in Sec 4.5. The visualization analysis can be found in Sec 4.6.

4.1. Datasets

We evaluated PIC with four DG benchmarks. **(1) OfficeHome** [43]. The OfficeHome dataset is also an object recognition dataset, which contains four domains: Art (A), Clipart (C), Product (P), and Real-World (R), including 65 categories of every domain. **(2) TerraIncognita**. The TerraIncognita dataset was proposed by Beery et al. [44]. Unlike previous datasets where images were acquired manually, the images in the TerraIncognita were captured automatically by fixed cameras and without human bias. The TerraIncognita dataset is divided into four domains: L100, L38, L43, and L46, and ten categories. **(3) PACS** [41]. PACS is a challenging benchmark dataset, which contains data from Photo (P), Art (A), Cartoon (C), and Sketch (S). The differences between the domains of PACS are obvious. In each domain, there are 7 categories. **(4) VLCS** [42]. The VLCS dataset was proposed to test the algorithm's ability to eliminate differences between domains. VLCS uses four standard datasets: Caltech101 (C), VOC2007 (V), LabelMe (L), and SUN09 (S) and contains images from 5 classes. The number of samples in the five categories is not balanced. For example, the number of samples for the person category is 6 times that of the dog category.

4.2. Experiments Details

Following the settings of DomainBed [45], we use ResNet-50 [46], pre-trained on ImageNet [47], as the backbone network. A fully-connected layer as the information disentanglement network, and a 3-layer multi-layer perceptron as the classification network and discriminator network. For each experimental result in PIC, we conduct 3 independent experiments with different random seeds and report the average accuracy to reduce the instability of the results. For the EPIC, we average the models from multiple independent experiments, hence there is no standard deviation term for the results of EPIC.

For data partitioning and model selection, we follow the experimental settings of DomainBed and split the source domain data into 80% training data and 20% validation data, and the validation data is used for hyperparameter selection. The model does not have access to the target domain data during the entire training process.

In terms of path setting, the number of paths that can be constructed increases exponentially with the number of source domains, for example, three source domains can form $3! = 6$ different paths. Therefore, it is impossible to use all paths for training. In practical experiments, we choose the same number of paths as the number of source domains and use each source domain as the endpoint of a path. Taking the PACS dataset as an example, if S is the target domain and $\{P, A, C\}$ are the source domains, we construct three paths: $P \rightarrow A \rightarrow C$, $A \rightarrow C \rightarrow P$, and $C \rightarrow P \rightarrow A$. We save the model with the best performance on the validation set on each path, and finally, average the models as the model used for testing.

We use SGD [48] as the optimizer during training, where the momentum is 0.9 and the initial learning rate is explored in the range of $[10^{-3.5}, 10^{-2.5}]$, with the learning rate decaying exponentially as the training step increases. The experimental results are compared with other methods on four publicly available datasets introduced in Sec 4.1. In fairness, we compare the results with those presented in DomainBed [45], which are the results after careful parameter tuning.

Table 1: Domain Generalization results on OfficeHome benchmark. White backgrounds are non-ensembled methods, gray backgrounds are ensembled methods. The best accuracies are indicated in bold and the second-best ones are underlined.

Method	A	C	P	R	Avg
ERM [49]	61.3 ± 0.7	52.4 ± 0.3	75.8 ± 0.1	76.6 ± 0.3	66.5
CORAL [23]	<u>65.3</u> ± 0.4	<u>54.4</u> ± 0.5	76.5 ± 0.1	78.4 ± 0.5	<u>68.7</u>
MMD [7]	60.4 ± 0.2	53.3 ± 0.3	74.3 ± 0.1	77.4 ± 0.6	66.3
DANN [10]	59.9 ± 1.3	53.0 ± 0.3	73.6 ± 0.7	76.9 ± 0.5	65.9
CDANN [11]	61.5 ± 1.4	50.4 ± 2.4	74.4 ± 0.9	76.6 ± 0.8	65.8
RSC [8]	60.7 ± 1.4	51.4 ± 0.3	74.8 ± 1.1	75.1 ± 1.3	65.5
SelfReg [50]	63.6 ± 1.4	53.1 ± 1.0	<u>76.9</u> ± 0.4	78.1 ± 0.4	67.9
Fishr [51]	63.4 ± 0.8	54.2 ± 0.3	76.4 ± 0.3	<u>78.5</u> ± 0.2	68.2
PIC (ours)	68.0 ± 0.6	58.5 ± 0.4	78.9 ± 0.4	80.6 ± 0.8	71.5
SWAD [12]	66.1 ± 0.4	57.7 ± 0.4	78.4 ± 0.1	80.2 ± 0.2	70.6
AdaClust [52]	-	-	-	-	69.4
EoA [13]	<u>69.1</u>	<u>59.8</u>	<u>79.5</u>	<u>81.5</u>	<u>72.5</u>
EPIC (ours)	69.5	60.4	80.4	82.0	73.1

4.3. Results

The experiments were conducted on four test data sets of OfficeHome, TerraIncognita, PACS, and VLCS. The results of one single dataset are shown in Table 1, Table 2, Table 3, and Table 4 respectively, and the average experimental results on the four datasets are shown in Table 5. The experimental results are compared with 11 domain generalization methods, ERM [49], CORAL [23], MMD [7], DANN [10], CDANN [11], RSC [8], SelfReg [50], Fishr [51], SWAD [12], AdaClust(with SWAD) [52] and EoA [13]. They cover the main current domain generalization research solutions.

4.4. Results Analysis

In this section, we evaluate and analyze the results of our experiments. For a fair comparison, all models use the same backbone, data partitioning, and model selection methods as suggested by DomainBed.

OfficeHome benchmark. Our method obtains outstanding results in the OfficeHome dataset, and our method surpasses other methods in every task. In particular, the average accuracy of PIC and EPIC was 71.5% and 73.1%, correspondingly. With the same evaluation protocol, PIC and EPIC have 2.8%

Table 2: Domain Generalization results on TerraIncognita benchmark. White backgrounds are non-ensembled methods, gray backgrounds are ensembled methods. The best accuracies are indicated in bold and the second-best ones are underlined.

Method	L100	L38	L43	L46	Avg
ERM [49]	49.8 ± 4.4	42.1 ± 1.4	56.9 ± 1.8	35.7 ± 3.9	46.1
CORAL [23]	<u>51.6</u> ± 2.4	42.2 ± 1.0	57.0 ± 1.0	39.8 ± 2.9	<u>47.6</u>
MMD [7]	41.9 ± 3.0	34.8 ± 1.0	57.0 ± 1.9	35.2 ± 1.8	42.2
DANN [10]	51.1 ± 3.5	40.6 ± 0.6	57.4 ± 0.5	37.7 ± 1.8	46.7
CDANN [11]	47.0 ± 1.9	41.3 ± 4.8	54.9 ± 1.7	39.8 ± 2.3	45.8
RSC [8]	50.2 ± 2.2	39.2 ± 1.4	56.3 ± 1.4	40.8 ± 0.6	46.6
SelfReg [50]	48.8 ± 0.9	41.3 ± 1.8	<u>57.3</u> ± 0.7	<u>40.6</u> ± 0.9	47.0
Fishr [51]	50.2 ± 3.9	43.9 ± 0.8	55.7 ± 2.2	39.8 ± 1.0	47.4
PIC (ours)	58.0 ± 0.6	<u>43.8</u> ± 0.4	53.0 ± 0.4	38.7 ± 0.8	48.4
SWAD [12]	55.4 ± 0.0	44.9 ± 1.1	<u>59.7</u> ± 0.4	39.9 ± 0.2	50.0
AdaClust [52]	-	-	-	-	50.6
EoA [13]	<u>57.8</u>	46.5	61.3	43.5	52.3
EPIC (ours)	58.0	48.9	58.0	<u>42.2</u>	<u>51.8</u>

and 0.6% improvement over the best competitor, respectively. The brilliant results in different domains of OfficeHome demonstrate the effectiveness of our method.

TerraIncognita benchmark. The samples of the TerraIncognita dataset consist of an empty category and 9 wildlife species. Samples are collected by fixed cameras without human intervention. Thus the illumination and object positions in the data set have diverse variations, which challenges the robustness of the model to data changes. Our method has good performance on this dataset. PIC achieves optimal results among algorithms without ensemble learning, and EPIC achieves sub-optimal results among algorithms with ensemble learning. Experimental results show the good robustness of our method.

PACS benchmark. Our method achieves competitive results on the PACS dataset. PIC achieves optimal results in one task. EPIC achieves optimal results in two tasks. In particular, EPIC achieves a 3% improvement over other methods in the Sketch domain, which is composed of sparse feature distribution and is difficult to learn.

VLCS benchmark. On the VLCS dataset, both PIC and EPIC obtained the second-best results, outperforming most

Table 3: Domain Generalization results on PACS benchmark. White backgrounds are non-ensembled methods, gray backgrounds are ensembled methods. The best accuracies are indicated in bold and the second-best ones are underlined.

Method	A	C	P	S	Avg
ERM [49]	84.7 $\pm$ 0.4	<u>80.8</u> $\pm$ 0.6	97.2 $\pm$ 0.3	79.3 $\pm$ 1.0	85.5
CORAL [23]	88.3 $\pm$ 0.2	80.0 $\pm$ 0.5	97.5 $\pm$ 0.3	78.8 $\pm$ 1.3	<u>86.2</u>
MMD [7]	86.1 $\pm$ 1.4	79.4 $\pm$ 0.9	96.6 $\pm$ 0.2	76.5 $\pm$ 0.5	84.6
DANN [10]	86.4 $\pm$ 0.8	77.4 $\pm$ 0.8	97.3 $\pm$ 0.4	73.5 $\pm$ 2.3	83.6
CDANN [11]	84.6 $\pm$ 1.8	75.5 $\pm$ 0.9	96.8 $\pm$ 0.3	73.5 $\pm$ 0.6	82.6
RSC [8]	85.4 $\pm$ 0.8	79.7 $\pm$ 1.8	<u>97.6</u> $\pm$ 0.3	78.2 $\pm$ 1.2	85.2
SelfReg [50]	<u>87.9</u> $\pm$ 1.0	79.4 $\pm$ 1.4	96.8 $\pm$ 0.7	78.3 $\pm$ 1.2	85.6
Fishr [51]	<u>87.9</u> $\pm$ 0.6	<u>80.8</u> $\pm$ 0.5	97.9 $\pm$ 0.4	81.1 $\pm$ 0.8	86.9
PIC (ours)	86.2 $\pm$ 0.3	81.3 $\pm$ 0.2	96.3 $\pm$ 0.3	<u>80.4</u> $\pm$ 0.2	86.1
SWAD [12]	<u>89.3</u> $\pm$ 0.2	<u>83.4</u> $\pm$ 0.6	<u>97.3</u> $\pm$ 0.3	<u>82.5</u> $\pm$ 0.5	88.1
AdaClust [52]	-	-	-	-	89.2
EoA [13]	90.5	<u>83.4</u>	98.0	<u>82.5</u>	<u>88.6</u>
EPIC (ours)	87.4	83.5	97.1	85.5	88.4

methods. Since the samples in the VLCS vary widely in number and distribution, our method obtains competitive results, which indicates the stable feature extraction capability of the method.

Overall result analysis. A total of 16 learning tasks are constructed in the four test benchmarks, and our method achieved optimal results in more than half of the tasks, as shown in the Table 5. The average performance of PIC and EPIC is 71.2% and 73.2%, outperforming their respective competitors, which demonstrates the rationality and robustness of our method.

4.5. Ablation study

In this section, we perform ablation studies to investigate the effectiveness of the designed stepwise learning method and the proposed path information constraint. The ablation study is conducted on PACS and the results can be found in Table 6.

Analysis for Stepwise learning on one path. The stepwise learning method gradually loads data in different fields on the data path, and has a Gradually Vanishing Bridge (GVB) auxiliary learning structure that enables feature decoupling via adversarial learning to reduce the impact of domain-dependent features on generalization. As shown in the Table 6, this method has a certain improvement compared with ERM.

Table 4: Domain Generalization results on VLCS benchmark. White backgrounds are non-ensembled methods, gray backgrounds are ensembled methods. The best accuracies are indicated in bold and the second-best ones are underlined.

Method	C	L	S	V	Avg
ERM [49]	97.7 $\pm$ 0.4	64.3 $\pm$ 0.9	<u>73.4</u> $\pm$ 0.5	74.6 $\pm$ 1.3	77.5
CORAL [23]	98.3 $\pm$ 0.1	<u>66.1</u> $\pm$ 1.2	<u>73.4</u> $\pm$ 0.3	<u>77.5</u> $\pm$ 1.2	78.8
MMD [7]	97.7 $\pm$ 0.1	64.0 $\pm$ 1.1	72.8 $\pm$ 0.2	75.3 $\pm$ 3.3	77.5
DANN [10]	99.0 $\pm$ 0.3	65.1 $\pm$ 1.4	73.1 $\pm$ 0.3	77.2 $\pm$ 0.6	<u>78.6</u>
CDANN [11]	97.1 $\pm$ 0.3	65.1 $\pm$ 1.2	70.7 $\pm$ 0.8	77.1 $\pm$ 1.5	77.5
RSC [8]	97.9 $\pm$ 0.1	62.5 $\pm$ 0.7	72.3 $\pm$ 1.2	75.6 $\pm$ 0.8	77.1
SelfReg [50]	96.7 $\pm$ 0.4	65.2 $\pm$ 1.2	73.1 $\pm$ 1.3	76.2 $\pm$ 0.7	77.8
Fishr [51]	97.6 $\pm$ 0.7	67.3 $\pm$ 0.5	72.2 $\pm$ 0.9	75.7 $\pm$ 0.3	78.2
PIC (ours)	<u>98.4</u> $\pm$ 0.2	61.3 $\pm$ 0.6	77.7 $\pm$ 1.0	77.6 $\pm$ 0.4	78.8
SWAD [12]	98.4 $\pm$ 0.1	61.3 $\pm$ 0.3	77.7 $\pm$ 0.5	<u>77.6</u> $\pm$ 0.6	78.8
AdaClust [52]	-	-	-	-	79.6
EoA [13]	99.1	<u>63.1</u>	75.9	78.3	79.1
EPIC (ours)	<u>98.9</u>	65.1	<u>76.8</u>	76.8	<u>79.4</u>

Table 5: Domain Generalization average results. White backgrounds are non-ensembled methods, gray backgrounds are ensembled methods. The best accuracies are indicated in bold and the second best-ones are underlined.

Method	VLCS	PACS	OfficeHome	TerraIncognita	Avg
ERM [49]	77.5	85.5	66.5	46.1	68.9
CORAL [23]	78.8	<u>86.2</u>	<u>68.7</u>	<u>47.6</u>	<u>70.3</u>
MMD [7]	77.5	84.6	66.3	42.2	67.7
DANN [10]	<u>78.6</u>	83.6	65.9	46.7	68.7
CDANN [11]	77.5	82.6	65.8	45.8	67.9
RSC [8]	77.1	85.2	65.5	46.6	68.6
SelfReg [50]	77.8	85.6	67.9	47.0	69.6
Fishr [51]	78.2	86.9	68.2	47.4	70.2
PIC (ours)	78.8	86.1	71.5	48.4	71.2
SWAD [12]	79.1	88.1	70.6	50.0	72.0
AdaClust [52]	79.6	89.2	69.4	50.6	72.2
EoA [13]	79.1	<u>88.6</u>	<u>72.5</u>	52.3	<u>73.1</u>
EPIC (ours)	<u>79.4</u>	88.4	73.1	<u>51.8</u>	73.2

Table 6: Ablation study on OfficeHome.

Method	A	C	P	R	Avg
ERM	61.3	52.4	75.8	76.6	66.5
Stepwise	66.3	57.0	77.9	80.3	70.4
PIC	68.0	58.5	78.9	80.6	71.5
EPIC	69.5	60.4	80.4	82.0	73.1

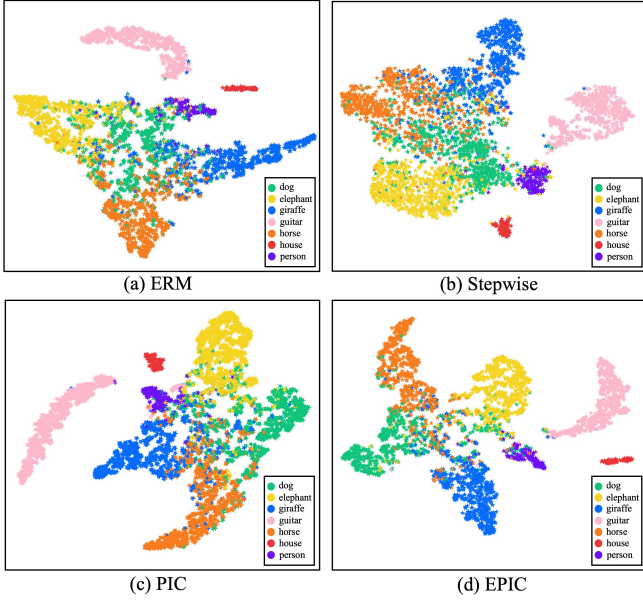


Figure 4: t-SNE visualization on ERM and our method. 7 colors represent each of the 7 different categories in the PACS benchmark. The source domain in the experiment is art, photo, and cartoon. What is visualized here is the testing result on the target sketch domain.

Analysis for Path Information Constraint. The ablation study experiment for path information constraint is to compare PIC with the stepwise learning method without using path constraints. As shown in Table 6, with the path information constraint, our method results in higher accuracy on the testing set. The further improvement in accuracy demonstrates the effectiveness of the path information constraint in learning generalizable representations. By applying ensemble learning methods, the effectiveness of PIC is further demonstrated.

4.6. Representations Visualization

We visualize the learned representations of the target domain sketch in the PACS benchmark. As shown in Figure 4(a), the classification boundary of the ERM algorithm in the target domain is blurred, where the dog category has a large overlap with other categories without forming a clear classification boundary. The designed stepwise learning method result is shown in Figure 4(b). Although the dog category still has a wide range of distribution, the overlap area with other species is smaller.

The result of PIC is shown in Figure 4(c). The dog category is more concentrated in the right-center area, and other categories

are also more concentrated. Furthermore, the number of confusing data is reduced, and the classification boundary is clear. The result of EPIC is shown in Figure 4(d), the aliasing area is very small, and the classification boundaries become apparent. The visualization results are consistent with the experimental results, further verifying that PIC has a strong learning ability for robust features and that ensemble learning has a strong auxiliary improvement ability for path constraints.

5. Conclusion

In this paper, we propose a simple and effective framework for domain generalization. It uses path information constraint (PIC) to deal with the problem that most existing domain generalization methods ignore the information brought by the variation of source domain data distribution. PIC utilizes the data distribution changes in the source domain to construct multiple learning paths to realize diversified exploration in the training process. Experiments on four test benchmarks obtained competitive results. In particular, considerable improvements are achieved on the challenging OfficeHome dataset. These experimental results validate the effectiveness of our method.

As for the shortcomings of the work, it is prone to prove that as the number of source domains increases, the number of possible paths grows explosively. Therefore, we artificially select the learning paths in practical experiments. We believe that it would be promising future work if an algorithm could be designed to search for the suitable path automatically instead of our manual operation.

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